

x5bp3uydu

June 21, 2024

How Do the Games' Genres, their Sales and User and Critics Score, and Series, Focus of the games connected with the Game's Global Sales

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June 2024

##Abstract This study carried out to find the association between the influence of factors such as the genre of the game, the evaluation of the game by users, the critics' evaluation, is game a part of a series of games, what is the main focus of the game on their expected global sales. A multiple regression model has been constructed to address this problem. In the course of the work, it turned out that user score, critics score and series are significant predictors of global sales. Ultimately, the constructed model can be used as a tool for forecasting sales of games on the global market.

0.1 Introduction

Currently, there are many games of various genres, and this list is constantly being updated. Each genre demonstrates a completely different gameplay, requires different skills and abilities. There are many explanations why certain games are liked by different groups of society, with different interests, different genders and ages. Of course, this gives reason to believe that the genre can predict the success of this game.

Moreover, critical or user ratings can be important aspects for predicting the level of global sales of games. Critics often give ratings before the start of game sales, which affects the very first impression of users who have not tried the gameplay yet, but have already heard about this game. However, given that not everyone is ready to trust critics, the evaluation of users is also important. If the game is a series of games, then it may seem that sales success will be higher due to recognition, which we found out in our work. And it was also important to find out which type of game would bring more success to developers based on gameplay or narrative.

In this work, we are trying to figure out and give an accurate answer to the following question: "Does the genre of the game, the evaluation of users, the evaluation of critics, the factor of whether the game is a series of games, and what is the focus of the game, on its sales and evaluation by users and experts? And if so, to what extent?". To answer this question, a multiple linear regression model based on data from the Tagged-Data-Final.csv dataset was used. The task of this study is to find a dependency, if any, so that game creators and publishers can have information about its sales before choosing a game genre. This can help companies and people involved in game development to improve the quality of their work.

The dependent variables are `User_Score`, `Critics_Score`, `Genre`, `StoryFocus` or `GameplayFocus`, `Series`.

The fitted multiple linear regression model explains which of these variables are related and affect global game sales. More details regarding the model can be found in the results, and their meaning is discussed in the conclusion. The model that has been constructed is limited by the time the data was collected.

###Hypotheses **Null Hypothesis (Ho):** The variables (`User_Score`, `Critics_Score`, `Genre`, `StoryFocus` or `GameplayFocus`, `Series`) do not affect the sales or user and expert evaluation of games.
H0: `User_Score = Critics_Score = Genre = StoryFocus or GameplayFocus = Series = 0`

Alternative Hypothesis (Ha): The variables (`User_Score`, `Critics_Score`, `Genre`, `StoryFocus` or `GameplayFocus`, `Series`) affect the sales or user and expert evaluation of games.

Ha: `User_Score = Critics_Score = Genre = StoryFocus or GameplayFocus = Series = 0`

##Data A very suitable dataset was found in the file *Tagged-Data-Final.csv*, which was uploaded by the user *The Devastator* on [kaggle.com](https://www.kaggle.com).

The original (raw) dataset consists of 11563 data lines representing 16 columns, and the final dataset consists of 4424 data lines, representing 16 columns. It collected data about games' sales, genre, publisher, platform, rating, story focus and gameplay focus for each game by 2016.

The DataFrame `games` contains data on 6894 games with 18 variables (columns):

- *Name*: Name of the game
- *Year_of_Release*: Year of release
- *Genre*: The genre of the game
- *Publisher*: Publisher of the game
- *NA_Sales*: Sales in North America (in millions of copies)
- *EU_Sales*: Sales in Europe (in millions of copies)
- *JP_Sales*: Sales in Japan (in millions of copies)
- *Other_Sales*: Sales in other regions (in millions of copies)
- *Global_Sales*: Global sales (in millions of copies)
- *Critic_Score*: Average critics score (on a 100-point scale)
- *Critic_Count*: The number of critics' ratings
- *User_Score*: Average user rating (on a 10-point scale)
- *User_Count*: The number of user ratings
- *Developer*: Game Developer
- *Rating*: Age rating of the game
- *StoryFocus*: Does the game focus on the plot (values 'x' = [YES] or NaN = [NO])
- *Gameplay Focus*: Does the game focus on gameplay (values 'x' = [YES] or NaN = [NO])
- *Series*: Is the game part of a series (values 'x' = [YES] or NaN = [NO])

###Population People who play games, being calculated as split to the regions they lived in 2016, and globally at the same time. You can generalize the answers to all the people who play games. For greater reliability, in the future, it is advised to re-analyze (starting with data collection) to ensure that the data does not lose its relevance over time.

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```

from scipy import stats
from scipy.stats import f_oneway, probplot

import statsmodels.api as sm
from statsmodels.formula.api import ols
import numpy as np

sns.set_style("whitegrid")

```

```

[2]: # needed/convenient only if you're working in Google colab (not for local
      ↪ Jupyter)
from google.colab import files
uploaded = files.upload()

```

<IPython.core.display.HTML object>

Saving Tagged-Data-Final.csv to Tagged-Data-Final.csv

```

[3]: games = pd.read_csv("Tagged-Data-Final.csv")
games

```

```

[3]:

```

	Name	Year_of_Release	Genre	\		
0	.hack//Infection Part 1	2002.0	Role-Playing			
1	.hack//Mutation Part 2	2002.0	Role-Playing			
2	.hack//Outbreak Part 3	2002.0	Role-Playing			
3	[Prototype]	2009.0	Action			
4	[Prototype]	2009.0	Action			
...	...	...	...			
6889	Zubo	2008.0	Misc			
6890	Zumba Fitness	2010.0	Sports			
6891	Zumba Fitness: World Party	2013.0	Misc			
6892	Zumba Fitness Core	2012.0	Misc			
6893	Zumba Fitness Rush	2012.0	Sports			
	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	\
0	Atari	0.49	0.38	0.26	0.13	
1	Atari	0.23	0.18	0.20	0.06	
2	Atari	0.14	0.11	0.17	0.04	
3	Activision	0.84	0.35	0.00	0.12	
4	Activision	0.65	0.40	0.00	0.19	
...	...	...	...	...	...	
6889	Electronic Arts	0.08	0.02	0.00	0.01	
6890	505 Games	1.74	0.45	0.00	0.18	
6891	Majesco Entertainment	0.17	0.05	0.00	0.02	
6892	505 Games	0.00	0.05	0.00	0.00	
6893	505 Games	0.00	0.16	0.00	0.02	
	Global_Sales	Critic_Score	Critic_Count	User_Score	User_Count	\

0	1.27	75.0	35.0	8.5	60.0
1	0.68	76.0	24.0	8.9	81.0
2	0.46	70.0	23.0	8.7	19.0
3	1.31	78.0	83.0	7.8	356.0
4	1.24	79.0	53.0	7.7	308.0
...	...	...	...	...	...
6889	0.11	75.0	19.0	7.6	75.0
6890	2.37	42.0	10.0	5.5	16.0
6891	0.24	73.0	5.0	6.2	40.0
6892	0.05	77.0	6.0	6.7	6.0
6893	0.18	73.0	7.0	6.2	5.0

	Developer	Rating	Story Focus	Gameplay Focus	Series
0	CyberConnect2	T	x	NaN	x
1	CyberConnect2	T	x	NaN	x
2	CyberConnect2	T	x	NaN	x
3	Radical Entertainment	M	NaN	x	x
4	Radical Entertainment	M	NaN	x	x
...	...	...	...	...	...
6889	EA Bright Light	E10+	NaN	x	NaN
6890	Pipeworks Software, Inc.	E	NaN	x	NaN
6891	Zoe Mode	E	NaN	x	NaN
6892	Zoe Mode	E10+	NaN	x	NaN
6893	Majesco Games, Majesco	E10+	NaN	x	NaN

[6894 rows x 18 columns]

###The response variable 1. *Global Sales* – The sales of the game in the world,

###Explanatory variables

1. *User_Score* - Average score given by users to the game, given as a float
2. *Critics_Score* - Average score given by critics to the game, given as a float
3. *Genre* - Genre the game belongs to, given as a String
4. *Story Focus* - Does the game focuses on story , values - 'x' = [YES] or NaN = [NO]

Important note: (in the code was changed to the [Story_Focus] to be int(1 for Yes, 0 for No))
(Mutually exclusive with [Gameplay Focus] and its' representations) 5. *Gameplay Focus* - Does the game focuses on gameplay , values - 'x' = [YES] or NaN = [NO]

Important note: (in the code was changed to the [Gameplay_Focus] to be int(1 for Yes, 0 for No))
(Mutually exclusive with [Story Focus] and its' representations) 6. *Series* - Is the game part of the series or it is independent, values - 'x' = [YES] or NaN = [NO]

Important note: (In the code was changed to be int(1 for Yes, 0 for No))

###Model

###Data preprocessing

```
[4]: # Handle missing values in relevant columns
games.dropna(subset=['Global_Sales', 'Critic_Score', 'User_Score'],
             inplace=True)

# Convert 'Series', 'Story Focus', 'Gameplay Focus' columns to the int values
games['Series'] = games['Series'].apply(lambda x: 1 if pd.notna(x) and x == 'x'
                                       else 0)
games['Story Focus'] = games['Story Focus'].apply(lambda x: 1 if pd.notna(x)
                                       and x == 'x' else 0)
games['Gameplay Focus'] = games['Gameplay Focus'].apply(lambda x: 1 if pd.
                                       notna(x) and x == 'x' else 0)

# Rename columns to avoid spaces and future problems
games.rename(columns={'Story Focus': 'Story_Focus', 'Gameplay Focus':
                    'Gameplay_Focus'}, inplace=True)

# Encode 'Genre' as dummy variables
games_encoded = pd.get_dummies(games, columns=['Genre'], drop_first=True)

games_encoded.columns = games_encoded.columns.str.replace(' ', '_').str.
                    replace('-', '_')

games_encoded #show data
```

```
[4]:
```

	Name	Year_of_Release	Publisher \
0	.hack//Infection Part 1	2002.0	Atari
1	.hack//Mutation Part 2	2002.0	Atari
2	.hack//Outbreak Part 3	2002.0	Atari
3	[Prototype]	2009.0	Activision
4	[Prototype]	2009.0	Activision
...	...	...	...
6889	Zubo	2008.0	Electronic Arts
6890	Zumba Fitness	2010.0	505 Games
6891	Zumba Fitness: World Party	2013.0	Majesco Entertainment
6892	Zumba Fitness Core	2012.0	505 Games
6893	Zumba Fitness Rush	2012.0	505 Games

	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales	Critic_Score \
0	0.49	0.38	0.26	0.13	1.27	75.0
1	0.23	0.18	0.20	0.06	0.68	76.0
2	0.14	0.11	0.17	0.04	0.46	70.0
3	0.84	0.35	0.00	0.12	1.31	78.0
4	0.65	0.40	0.00	0.19	1.24	79.0
...	...	...	...	...	...	...
6889	0.08	0.02	0.00	0.01	0.11	75.0
6890	1.74	0.45	0.00	0.18	2.37	42.0
6891	0.17	0.05	0.00	0.02	0.24	73.0

6892	0.00	0.05	0.00	0.00	0.05	77.0
6893	0.00	0.16	0.00	0.02	0.18	73.0

	Critic_Count	...	Genre_Fighting	Genre_Misc	Genre_Platform	\
0	35.0	...	False	False	False	
1	24.0	...	False	False	False	
2	23.0	...	False	False	False	
3	83.0	...	False	False	False	
4	53.0	...	False	False	False	
...	...	...	...	...	...	
6889	19.0	...	False	True	False	
6890	10.0	...	False	False	False	
6891	5.0	...	False	True	False	
6892	6.0	...	False	True	False	
6893	7.0	...	False	False	False	

	Genre_Puzzle	Genre_Racing	Genre_Role_Playing	Genre_Shooter	\
0	False	False	True	False	
1	False	False	True	False	
2	False	False	True	False	
3	False	False	False	False	
4	False	False	False	False	
...	...	...	...	...	
6889	False	False	False	False	False
6890	False	False	False	False	False
6891	False	False	False	False	False
6892	False	False	False	False	False
6893	False	False	False	False	False

	Genre_Simulation	Genre_Sports	Genre_Strategy
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False
...	...	...	...
6889	False	False	False
6890	False	True	False
6891	False	False	False
6892	False	False	False
6893	False	True	False

[6894 rows x 28 columns]

To start with, we had to remove the records that do not contain any values under the Global_Sales, Critic_Score and User_Score columns. This step is essential as it ensures the precision of our analysis by only considering complete data. Missing values can distort the results of the analysis and

lead to incorrect conclusions.

We will also convert categorical data into numeric data. The Series, Story Focus and Gameplay Focus columns contain the values 'x' for "Yes" and NaN for "No". We replace 'x' with 1 and NaN with 0 to simplify their use in the regression model, since regression models require numeric inputs. The column names were altered from Story Focus and Gameplay Focus to Story_Focus and Gameplay_Focus. This change was made with a view to ensuring smooth analysis and coding processes—having spaces in column names might lead to issues in error detection.

In the end, we will convert the categorical variable Genre into dummy variables. In simple terms, this action involves generating an individual column for every genre, with binary values: 0 or 1. Here, '1' denotes membership to that genre. Furthermore, by setting drop_first=True, we eliminate the initial genre column from our dataset in order to prevent multicollinearity.

###Multiple linear regression model

```
[5]: print(stats.ttest_ind(games['Global_Sales'], games['Critic_Score']))
      print(stats.ttest_ind(games['Global_Sales'], games['User_Score']))
      print(stats.ttest_ind(games['Global_Sales'], games['Series']))
      print(stats.ttest_ind(games['Global_Sales'], games['Story_Focus']))
      print(stats.ttest_ind(games['Global_Sales'], games['Gameplay_Focus' ]))
```

```
TtestResult(statistic=-412.15987253315325, pvalue=0.0, df=13786.0)
TtestResult(statistic=-219.31933301974675, pvalue=0.0, df=13786.0)
TtestResult(statistic=27.532136936041034, pvalue=1.7276650949912798e-162,
df=13786.0)
TtestResult(statistic=27.68759123823802, pvalue=2.943050054767639e-164,
df=13786.0)
TtestResult(statistic=-7.765289366903133, pvalue=8.718325447259215e-15,
df=13786.0)
```

```
[6]: # Assuming 'games_encoded' is your dataset with Global_Sales and other variables
      formula = 'Global_Sales ~ Critic_Score + User_Score + Series + '
      formula += ' + '.join([f'{col}' for col in games_encoded.columns if col.
      ↪startswith('Genre_') and col != 'Genre_Sports' and col != 'Genre_Puzzle' and
      ↪col != 'Genre_Simulation' and col != 'Genre_Fighting' and col !=
      ↪'Genre_Racing' and col != 'Genre_Shooter' and col != 'Genre_Platform'])
      # Genre_Sports, Gameplay_Focus, Genre_Puzzle, Genre_Simulation, Genre_Fighting,
      ↪Story_Focus, Genre_Racing, Genre_Shooter, Genre_Platform
      # Fit the initial model
      final_model = ols(formula, data=games_encoded).fit()

      # Print initial model summary
      print("Final Model Summary:")
      print(final_model.summary())
```

Final Model Summary:

OLS Regression Results

```
=====
Dep. Variable:          Global_Sales    R-squared:          0.070
```

```

Model:                OLS      Adj. R-squared:      0.069
Method:               Least Squares  F-statistic:      74.14
Date:                Fri, 21 Jun 2024  Prob (F-statistic):  6.11e-104
Time:                13:46:38    Log-Likelihood:    -14152.
No. Observations:    6894      AIC:                2.832e+04
Df Residuals:        6886      BIC:                2.837e+04
Df Model:             7
Covariance Type:      nonrobust

```

```

=====
=====

```

		coef	std err	t	P> t
[0.025	0.975]				

Intercept		-1.2665	0.132	-9.594	0.000
-1.525	-1.008				
Genre_Adventure[T.True]		-0.3435	0.119	-2.881	0.004
-0.577	-0.110				
Genre_Misc[T.True]		0.3835	0.100	3.840	0.000
0.188	0.579				
Genre_Role_Playing[T.True]		-0.1985	0.076	-2.614	0.009
-0.347	-0.050				
Genre_Strategy[T.True]		-0.6227	0.116	-5.354	0.000
-0.851	-0.395				
Critic_Score		0.0381	0.002	18.623	0.000
0.034	0.042				
User_Score		-0.0890	0.019	-4.572	0.000
-0.127	-0.051				
Series		0.3099	0.073	4.229	0.000
0.166	0.454				
=====					
Omnibus:	14313.426	Durbin-Watson:	1.508		
Prob(Omnibus):	0.000	Jarque-Bera (JB):	95688208.264		
Skew:	17.324	Prob(JB):	0.00		
Kurtosis:	579.123	Cond. No.	427.		
=====					

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

We used a multiple linear regression model where Global_Sales is a dependent variable (which we are trying to predict) and Critic_Score, User_Score, Genre, Series, Story_Focus and Gameplay_Focus are independent variables (factors that can affect sales). These factors were chosen to test the hypothesis that critical and user ratings, game genre, series affiliation, and focus on gameplay or storytelling can influence global sales of the game.

Using the OLS method, we find a linear relationship between a dependent variable and several independent variables.

That's what the output gives us:

R-squared (R^2): A low R-squared value indicates that there are many other factors that can affect sales, and they are not included in this model. The R-squared value: 0.070 shows that about 7% of the variation in global sales is explained by this model. This indicates that the model explains a small part of the variation in the data.

Coefficients and their significance: Critics' ratings and belonging to the series have a significant positive impact on game sales, while user ratings have a negative impact. The lack of significance of the focus on narrative or gameplay may indicate that these factors are not decisive for global sales.

Critic_Score: A coefficient of 0.0381 (p-value = 0.000) indicates a positive and significant impact of critics' ratings on sales.

User_Score: A coefficient of -0.0890 (p-value = 0.000) indicates a negative and significant impact of user ratings on sales.

Series: A coefficient of 0.3099 (p-value = 0.000) indicates a positive and significant impact of belonging to the series on sales.

Story_Focus and Gameplay_Focus: These variables do not have a significant impact on sales, since their p-value is > 0.05 .

Genres:

Adventure: A coefficient of -0.3435 (p-value = 0.000) indicates a negative and significant impact of on sales in case game is developed in it.

Fighting: This variable does not have a significant impact on sales, since its p-value is > 0.05 .

Platform: This variable does not have a significant impact on sales, since its p-value is > 0.05 .

Puzzle: This variable does not have a significant impact on sales, since its p-value is > 0.05 .

Racing: This variable does not have a significant impact on sales, since its p-value is > 0.05 .

Role-Playing: A coefficient of -0.1985 (p-value = 0.009) indicates a negative and significant impact of on sales in case game is developed in it.

Shooter: This variable does not have a significant impact on sales, since its p-value is > 0.05 .

Simulation: This variable does not have a significant impact on sales, since its p-value is > 0.05 .

Sports: This variable does not have a significant impact on sales, since its p-value is > 0.05 .

Strategy: A coefficient of -0.6227 (p-value = 0.000) indicates a negative and significant impact of on sales in case game is developed in it.

Misc: A coefficient of 0.3835 (p-value = 0.000) indicates a positive and significant impact of on sales in case game is developed in it.

```
[7]: # Correlation

correlation_results = {}
for genre in games['Genre'].unique():
    genre_data = games[games['Genre'] == genre]
    correlation = genre_data[['Global_Sales', 'Critic_Score', 'User_Score']].
    ↪corr()
    correlation_results[genre] = correlation
    print(f"Correlation for {genre}:\n", correlation)

# One-way ANOVA tests to check the mean difference
```

```

anova_sales = f_oneway(*(games[games['Genre'] == genre]['Global_Sales'] for
↳genre in games['Genre'].unique()))
print("ANOVA result for Global Sales by Genre:\n", anova_sales)

anova_critic = f_oneway(*(games[games['Genre'] == genre]['Critic_Score'] for
↳genre in games['Genre'].unique()))
print("ANOVA result for Critic Score by Genre:\n", anova_critic)

anova_user = f_oneway(*(games[games['Genre'] == genre]['User_Score'] for genre
↳in games['Genre'].unique()))
print("ANOVA result for User Score by Genre:\n", anova_user)

```

Correlation for Role-Playing:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.405149	0.125246
Critic_Score	0.405149	1.000000	0.501374
User_Score	0.125246	0.501374	1.000000

Correlation for Action:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.389324	0.193611
Critic_Score	0.389324	1.000000	0.606865
User_Score	0.193611	0.606865	1.000000

Correlation for Racing:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.245305	0.166397
Critic_Score	0.245305	1.000000	0.556247
User_Score	0.166397	0.556247	1.000000

Correlation for Misc:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.083161	0.076134
Critic_Score	0.083161	1.000000	0.612323
User_Score	0.076134	0.612323	1.000000

Correlation for Adventure:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.237371	0.096908
Critic_Score	0.237371	1.000000	0.723507
User_Score	0.096908	0.723507	1.000000

Correlation for Simulation:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.214531	0.102197
Critic_Score	0.214531	1.000000	0.620355
User_Score	0.102197	0.620355	1.000000

Correlation for Sports:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.110823	0.041338
Critic_Score	0.110823	1.000000	0.517810

User_Score	0.041338	0.517810	1.000000
------------	----------	----------	----------

Correlation for Shooter:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.349474	-0.034461
Critic_Score	0.349474	1.000000	0.563374
User_Score	-0.034461	0.563374	1.000000

Correlation for Puzzle:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.198505	0.148266
Critic_Score	0.198505	1.000000	0.734188
User_Score	0.148266	0.734188	1.000000

Correlation for Platform:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.280059	0.159374
Critic_Score	0.280059	1.000000	0.668235
User_Score	0.159374	0.668235	1.000000

Correlation for Strategy:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.273480	0.074707
Critic_Score	0.273480	1.000000	0.502022
User_Score	0.074707	0.502022	1.000000

Correlation for Fighting:

	Global_Sales	Critic_Score	User_Score
Global_Sales	1.000000	0.374771	0.256713
Critic_Score	0.374771	1.000000	0.645164
User_Score	0.256713	0.645164	1.000000

ANOVA result for Global Sales by Genre:

F_onewayResult(statistic=5.5388134299221, pvalue=6.875756411503829e-09)

ANOVA result for Critic Score by Genre:

F_onewayResult(statistic=18.377084916122264, pvalue=1.0070205794080673e-36)

ANOVA result for User Score by Genre:

F_onewayResult(statistic=10.548428673662386, pvalue=1.7150189045564263e-19)

As a result, we analyze the relationship between global sales, critics' ratings and user ratings in the context of various genres. Correlation analysis was used to find out how strongly sales and ratings of critics and user ratings are related. This helps to understand what factors can influence sales within each genre. ANOVA tests are used to check for significant differences in sales and ratings between different genres. We used it to determine whether genres affect ratings and sales, and how significant these differences are. That is, this analysis helps to draw conclusions about which genres are more successful in terms of sales and which factors influence the success of the game, depending on the genre.

For each genre, we get a correlation matrix showing how strongly global sales, critical ratings, and user ratings are related. A value of 1.000000 in the diagonal means an ideal correlation of the variable with itself. A value of 0.405149 indicates a moderate positive correlation between global sales and critical ratings. A value of 0.125246 indicates a weak positive correlation between global sales and user ratings.

Global Sales by Genre: A p value (6.875756411503829e-09) indicates that there is a statistically

significant difference in global sales between different genres

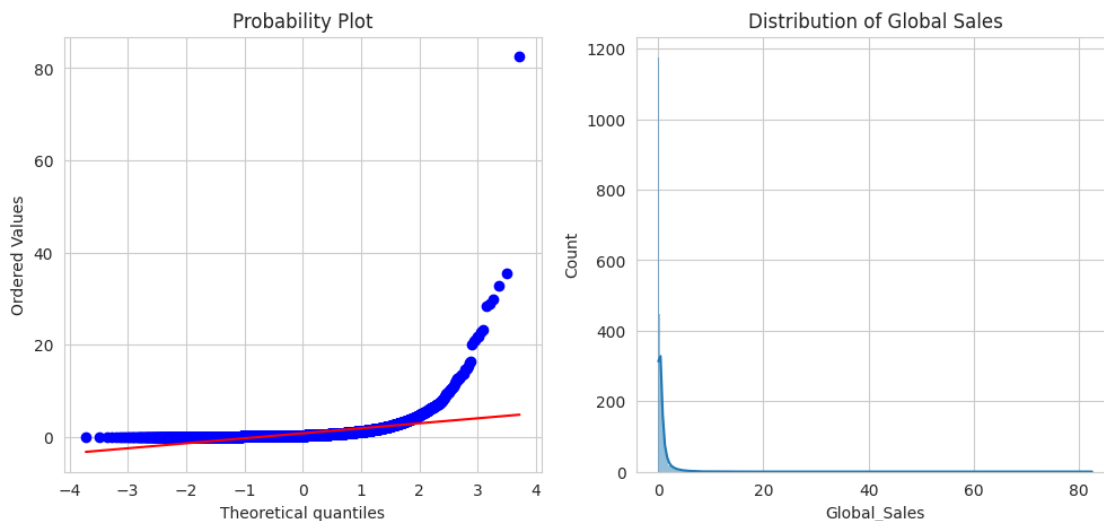
Critical Score by Genre: A p value (1.0070205794080673e-36) indicates that there is a statistically significant difference in critics' ratings between different genres.

User Score by Genre: The p value (1.7150189045564263e-19) indicates that there is a statistically significant difference in user ratings between different genres.

```
[8]: # Distribution of Global Sales
fig,ax = plt.subplots(1,2, figsize=(12,5))

stats.probplot(games['Global_Sales'], dist='norm', plot=ax[0])
plt.title('Probability Plot of Global Sales')
sns.histplot(games['Global_Sales'], kde=True)
plt.title('Distribution of Global Sales')
```

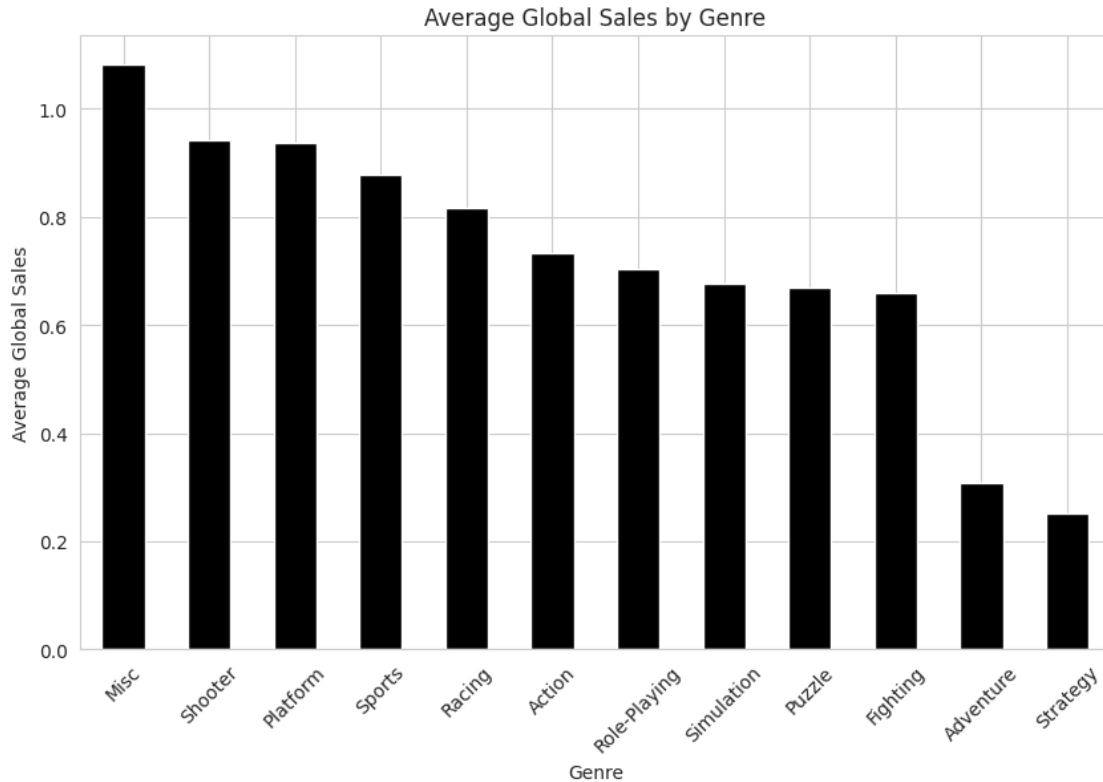
```
[8]: Text(0.5, 1.0, 'Distribution of Global Sales')
```



###Data visualization

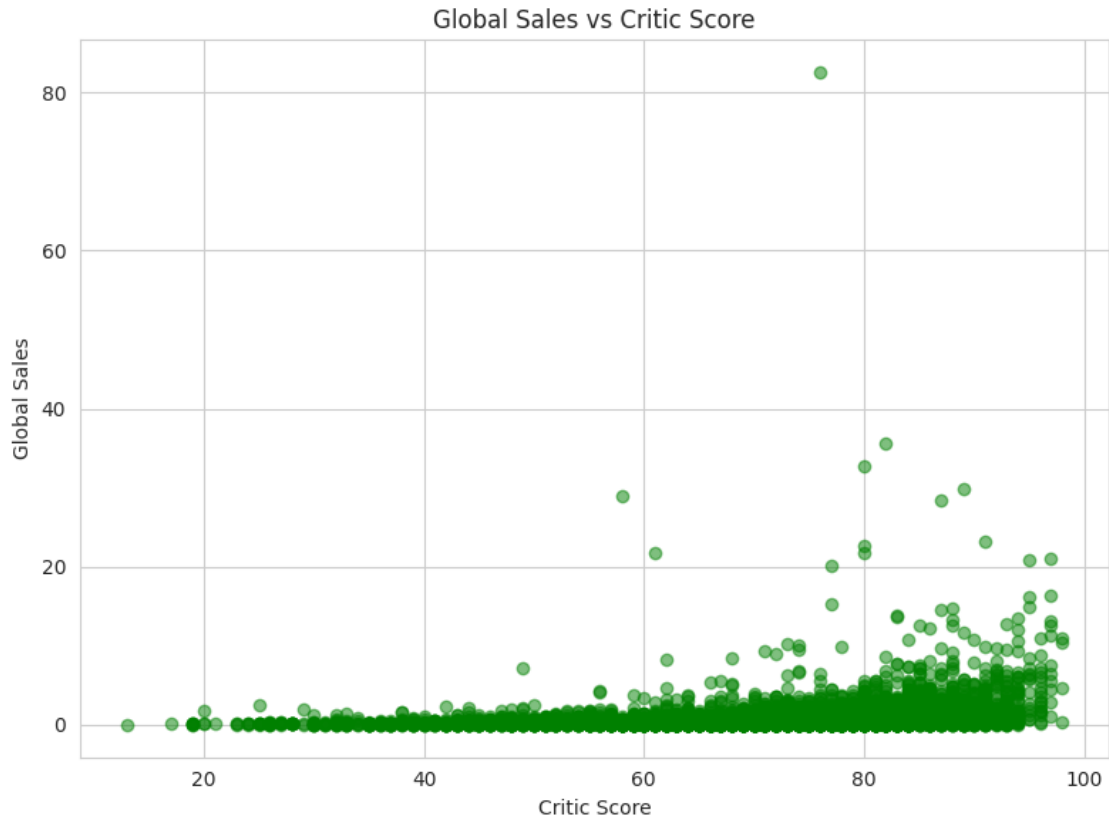
```
[9]: # Calculate mean global sales for each genre
mean_sales_by_genre = games.groupby('Genre')['Global_Sales'].mean()

# Bar plot for global sales by genre
plt.figure(figsize=(10, 6))
mean_sales_by_genre.sort_values(ascending=False).plot(kind='bar', color='black')
plt.title('Average Global Sales by Genre')
plt.xlabel('Genre')
plt.ylabel('Average Global Sales')
plt.xticks(rotation=45)
plt.show()
```



Next we'll look at the sales figures, for each genre shedding light on which genres tend to perform better overall. We've crafted a bar graph to visually represent the sales of games across different genres. To present this data effectively we've filtered it by genre computed the sales numbers for each genre and arranged them in descending order. This visual representation helps us identify which genres generally have sales figures providing insights into what factors contribute to the success of games. Upon inspection it's evident that in the coming months new game releases are predicted to achieve a level of 0.14. Following that you can explore genres like shooters, RPGs (role playing games) racing titles, platformers, miscellaneous games, fighting games, simulators, strategies and adventures. Later in our discussion is a puzzle game, with graphics expected for release in 0.02 years.

```
[10]: # Scatter plot [Global Sales vs Critic Score]
plt.figure(figsize=(8, 6))
plt.scatter(games_encoded['Critic_Score'], games_encoded['Global_Sales'],
            alpha=0.5, color='green')
plt.title('Global Sales vs Critic Score')
plt.xlabel('Critic Score')
plt.ylabel('Global Sales')
plt.grid(True)
plt.tight_layout()
plt.show()
```

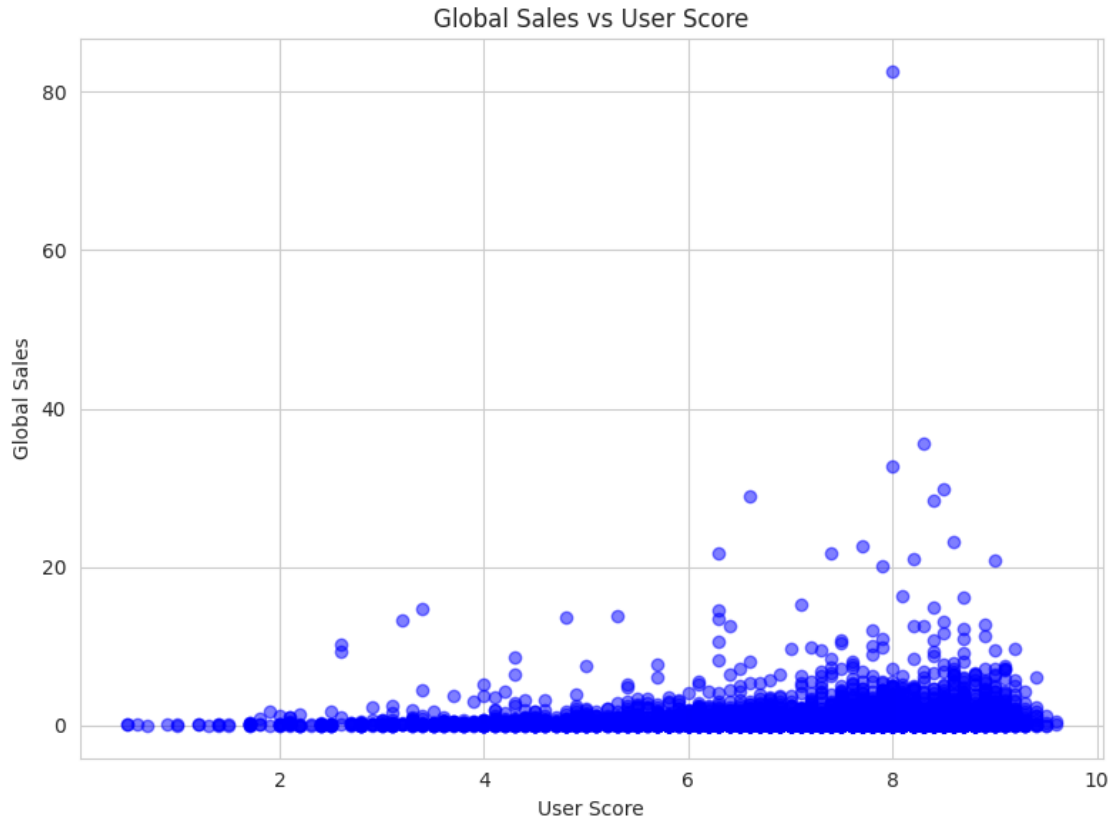


We created a scatter chart to visualize the relationship between global game sales and critical acclaim. The chart shows critical ratings on the X—axis, and global sales on the Y-axis. Thanks to the graph, we will see if there is a relationship between these variables.

The scatter plot shows the dependence of global game sales on critics' ratings. It can be seen that there is a positive correlation between these variables, that is, the higher the critics' score, the higher the global sales.

Before the estimates of 20, global sales were at 0 and after that they gradually increase. Reaching the mark of 100 according to critics, global sales clearly reach the level of 15 and vary by 15-35, moreover, there is an outlier, but only 1.

```
[11]: # Scatter plot [Global Sales vs User Score]
plt.figure(figsize=(8, 6))
plt.scatter(games_encoded['User_Score'], games_encoded['Global_Sales'], alpha=0.
↪5, color='blue')
plt.title('Global Sales vs User Score')
plt.xlabel('User Score')
plt.ylabel('Global Sales')
plt.grid(True)
plt.tight_layout()
plt.show()
```



We created a scatter chart to visualize the relationship between global game sales and user ratings. The graph on the X—axis has user ratings, and on the Y-axis there are global sales. This way we can find out if there is a relationship between these variables.

This graph shows how global game sales are related to user ratings. The points on the graph are located at level 0 on the X axis and begin to rise with increasing user ratings. At the rating level of 9 on the X axis, the points are at the global sales level of about 10. There are also less frequent points on the chart, varying at the global sales level of 10-30, and one outlier located far and high from the main mass of points.

```
[12]: # Doing regression analysis for each genre
regression_results = {}
for genre in games['Genre'].unique():
    genre_data = games[games['Genre'] == genre]
    if len(genre_data) > 1: # Ensure there's enough data for regression
        model = ols('Global_Sales ~ Critic_Score + User_Score',
data=genre_data).fit()
        regression_results[genre] = model.summary()
        print(f"Regression analysis for {genre}:\n", model.summary())
```

Regression analysis for Role-Playing:

OLS Regression Results

```

=====
Dep. Variable:          Global_Sales    R-squared:                0.172
Model:                  OLS             Adj. R-squared:           0.170
Method:                 Least Squares   F-statistic:              74.08
Date:                   Fri, 21 Jun 2024 Prob (F-statistic):       5.92e-30
Time:                   13:46:50        Log-Likelihood:           -1095.3
No. Observations:      715             AIC:                     2197.
Df Residuals:          712             BIC:                     2210.
Df Model:               2
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-1.8181	0.323	-5.628	0.000	-2.452	-1.184
Critic_Score	0.0470	0.004	11.605	0.000	0.039	0.055
User_Score	-0.1183	0.045	-2.640	0.008	-0.206	-0.030

```

=====
Omnibus:                534.273    Durbin-Watson:            1.282
Prob(Omnibus):          0.000      Jarque-Bera (JB):         7949.975
Skew:                   3.305      Prob(JB):                 0.00
Kurtosis:               17.938     Cond. No.                 573.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Action:

OLS Regression Results

```

=====
Dep. Variable:          Global_Sales    R-squared:                0.154
Model:                  OLS             Adj. R-squared:           0.153
Method:                 Least Squares   F-statistic:              149.9
Date:                   Fri, 21 Jun 2024 Prob (F-statistic):       1.64e-60
Time:                   13:46:50        Log-Likelihood:           -2827.9
No. Observations:      1644            AIC:                     5662.
Df Residuals:          1641            BIC:                     5678.
Df Model:               2
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-1.7695	0.187	-9.472	0.000	-2.136	-1.403
Critic_Score	0.0443	0.003	15.067	0.000	0.039	0.050
User_Score	-0.0712	0.030	-2.364	0.018	-0.130	-0.012

```

=====
Omnibus:                2055.840    Durbin-Watson:            1.397
Prob(Omnibus):          0.000      Jarque-Bera (JB):         350128.718
Skew:                   6.549      Prob(JB):                 0.00
=====

```

Kurtosis: 73.284 Cond. No. 392.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Racing:

OLS Regression Results

```
=====
Dep. Variable:      Global_Sales      R-squared:      0.061
Model:              OLS               Adj. R-squared:  0.058
Method:             Least Squares     F-statistic:    19.09
Date:               Fri, 21 Jun 2024   Prob (F-statistic): 9.29e-09
Time:               13:46:50          Log-Likelihood: -1285.5
No. Observations:   586              AIC:             2577.
Df Residuals:       583              BIC:             2590.
Df Model:           2
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-2.1034	0.508	-4.140	0.000	-3.101	-1.106
Critic_Score	0.0351	0.008	4.581	0.000	0.020	0.050
User_Score	0.0666	0.074	0.898	0.369	-0.079	0.212

```
=====
Omnibus:            952.846      Durbin-Watson:      0.903
Prob(Omnibus):      0.000      Jarque-Bera (JB):    389197.203
Skew:               9.590      Prob(JB):            0.00
Kurtosis:           127.788     Cond. No.            405.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Misc:

OLS Regression Results

```
=====
Dep. Variable:      Global_Sales      R-squared:      0.008
Model:              OLS               Adj. R-squared:  0.003
Method:             Least Squares     F-statistic:    1.531
Date:               Fri, 21 Jun 2024   Prob (F-statistic): 0.218
Time:               13:46:50          Log-Likelihood: -896.97
No. Observations:   386              AIC:             1800.
Df Residuals:       383              BIC:             1812.
Df Model:           2
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
--	------	---------	---	------	--------	--------

Intercept	-0.0276	0.647	-0.043	0.966	-1.300	1.245
Critic_Score	0.0098	0.011	0.908	0.364	-0.011	0.031
User_Score	0.0653	0.104	0.627	0.531	-0.140	0.270
=====						
Omnibus:		537.802	Durbin-Watson:			1.812
Prob(Omnibus):		0.000	Jarque-Bera (JB):			67103.200
Skew:		7.041	Prob(JB):			0.00
Kurtosis:		66.039	Cond. No.			356.
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Adventure:

OLS Regression Results

=====						
Dep. Variable:	Global_Sales	R-squared:				0.068
Model:	OLS	Adj. R-squared:				0.061
Method:	Least Squares	F-statistic:				9.536
Date:	Fri, 21 Jun 2024	Prob (F-statistic):				0.000101
Time:	13:46:50	Log-Likelihood:				-234.30
No. Observations:	264	AIC:				474.6
Df Residuals:	261	BIC:				485.3
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	-0.2657	0.183	-1.452	0.148	-0.626	0.095
Critic_Score	0.0150	0.004	4.055	0.000	0.008	0.022
User_Score	-0.0592	0.033	-1.814	0.071	-0.124	0.005
=====						
Omnibus:		300.019	Durbin-Watson:			1.332
Prob(Omnibus):		0.000	Jarque-Bera (JB):			12502.996
Skew:		4.894	Prob(JB):			0.00
Kurtosis:		35.262	Cond. No.			345.
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Simulation:

OLS Regression Results

=====			
Dep. Variable:	Global_Sales	R-squared:	0.048
Model:	OLS	Adj. R-squared:	0.041
Method:	Least Squares	F-statistic:	7.468

Date: Fri, 21 Jun 2024 Prob (F-statistic): 0.000684
Time: 13:46:50 Log-Likelihood: -483.49
No. Observations: 302 AIC: 973.0
Df Residuals: 299 BIC: 984.1
Df Model: 2
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-0.7825	0.413	-1.894	0.059	-1.595	0.031
Critic_Score	0.0248	0.007	3.414	0.001	0.011	0.039
User_Score	-0.0386	0.055	-0.698	0.486	-0.147	0.070
Omnibus:	341.308		Durbin-Watson:	1.225		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	15925.174		
Skew:	4.947		Prob(JB):	0.00		
Kurtosis:	37.171		Cond. No.	425.		

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Sports:

OLS Regression Results

Dep. Variable:	Global_Sales	R-squared:	0.013			
Model:	OLS	Adj. R-squared:	0.011			
Method:	Least Squares	F-statistic:	6.065			
Date:	Fri, 21 Jun 2024	Prob (F-statistic):	0.00241			
Time:	13:46:50	Log-Likelihood:	-2436.7			
No. Observations:	951	AIC:	4879.			
Df Residuals:	948	BIC:	4894.			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	-0.9350	0.623	-1.500	0.134	-2.159	0.288
Critic_Score	0.0290	0.009	3.239	0.001	0.011	0.047
User_Score	-0.0475	0.082	-0.581	0.561	-0.208	0.113
=====						
Omnibus:	2176.920	Durbin-Watson:	1.846			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	9766132.814			
Skew:	20.329	Prob(JB):	0.00			
Kurtosis:	497.784	Cond. No.	463.			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Shooter:

OLS Regression Results

Dep. Variable:	Global_Sales	R-squared:	0.201			
Model:	OLS	Adj. R-squared:	0.199			
Method:	Least Squares	F-statistic:	108.5			
Date:	Fri, 21 Jun 2024	Prob (F-statistic):	9.12e-43			
Time:	13:46:50	Log-Likelihood:	-1668.3			
No. Observations:	868	AIC:	3343.			
Df Residuals:	865	BIC:	3357.			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	-0.9901	0.315	-3.145	0.002	-1.608	-0.372
Critic_Score	0.0696	0.005	14.686	0.000	0.060	0.079
User_Score	-0.4242	0.046	-9.211	0.000	-0.515	-0.334
=====						
Omnibus:	699.904	Durbin-Watson:	1.408			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	14088.194			
Skew:	3.619	Prob(JB):	0.00			
Kurtosis:	21.361	Cond. No.	408.			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Puzzle:

OLS Regression Results

=====						
Dep. Variable:	Global_Sales	R-squared:	0.039			
Model:	OLS	Adj. R-squared:	0.023			
Method:	Least Squares	F-statistic:	2.360			
Date:	Fri, 21 Jun 2024	Prob (F-statistic):	0.0990			
Time:	13:46:50	Log-Likelihood:	-221.53			
No. Observations:	118	AIC:	449.1			
Df Residuals:	115	BIC:	457.4			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	-1.0211	0.860	-1.187	0.238	-2.725	0.683
Critic_Score	0.0232	0.016	1.445	0.151	-0.009	0.055
User Score	0.0067	0.163	0.041	0.968	-0.317	0.330

```
=====
Omnibus:                196.693    Durbin-Watson:                1.867
Prob(Omnibus):          0.000    Jarque-Bera (JB):          16966.922
Skew:                   6.786    Prob(JB):                  0.00
Kurtosis:               60.155    Cond. No.                  423.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Platform:

OLS Regression Results

```
=====
Dep. Variable:          Global_Sales    R-squared:                0.080
Model:                  OLS            Adj. R-squared:           0.075
Method:                 Least Squares   F-statistic:             17.35
Date:                   Fri, 21 Jun 2024 Prob (F-statistic):       5.94e-08
Time:                   13:46:50        Log-Likelihood:          -910.77
No. Observations:      403            AIC:                    1828.
Df Residuals:          400            BIC:                    1840.
Df Model:               2
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-2.5179	0.678	-3.712	0.000	-3.851	-1.185
Critic_Score	0.0583	0.012	4.864	0.000	0.035	0.082
User_Score	-0.0844	0.108	-0.778	0.437	-0.298	0.129

```
=====
Omnibus:                628.557    Durbin-Watson:                1.414
Prob(Omnibus):          0.000    Jarque-Bera (JB):          149054.726
Skew:                   8.508    Prob(JB):                  0.00
Kurtosis:               95.667    Cond. No.                  420.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Strategy:

OLS Regression Results

```
=====
Dep. Variable:          Global_Sales    R-squared:                0.080
Model:                  OLS            Adj. R-squared:           0.073
Method:                 Least Squares   F-statistic:             11.96
Date:                   Fri, 21 Jun 2024 Prob (F-statistic):       1.04e-05
Time:                   13:46:50        Log-Likelihood:          -160.75
No. Observations:      278            AIC:                    327.5
Df Residuals:          275            BIC:                    338.4
=====
```

Df Model: 2
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-0.4726	0.172	-2.746	0.006	-0.811	-0.134
Critic_Score	0.0123	0.003	4.717	0.000	0.007	0.017
User_Score	-0.0237	0.019	-1.251	0.212	-0.061	0.014
Omnibus:		322.520	Durbin-Watson:			1.809
Prob(Omnibus):		0.000	Jarque-Bera (JB):			19064.494
Skew:		4.953	Prob(JB):			0.00
Kurtosis:		42.341	Cond. No.			491.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression analysis for Fighting:

OLS Regression Results

Dep. Variable:	Global_Sales	R-squared:	0.141			
Model:	OLS	Adj. R-squared:	0.136			
Method:	Least Squares	F-statistic:	30.82			
Date:	Fri, 21 Jun 2024	Prob (F-statistic):	4.04e-13			
Time:	13:46:50	Log-Likelihood:	-535.72			
No. Observations:	379	AIC:	1077.			
Df Residuals:	376	BIC:	1089.			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	-1.3585	0.290	-4.688	0.000	-1.928	-0.789
Critic_Score	0.0269	0.005	5.727	0.000	0.018	0.036
User_Score	0.0195	0.048	0.409	0.683	-0.074	0.113
=====						
Omnibus:	470.443	Durbin-Watson:	1.296			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	46015.559			
Skew:	5.733	Prob(JB):	0.00			
Kurtosis:	55.749	Cond. No.	405.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The last step of the analysis involves performing regression analysis, for each type of video game

genre in the dataset. Specifically the analyst creates regression models to forecast sales based on both user ratings within each genre.

These findings illustrate how the ratings from critics and users impact the sales of games across genres. This insight helps determine whether critics reviews or user feedback has a influence on game sales, within various genres. By understanding the significance of ratings companies can prioritize enhancing aspects of games that matter most to either critics or users depending on the genre. This enables developers to target elements of games effectively to drive sales.

In most genres, critics' ratings have a significant positive impact on global sales. For example, in the Shooter genre, the Critic_Score coefficient = 0.0696 ($p < 0.000$), which indicates a high significant impact.

The impact of user ratings is less clear. In some genres, such as Role-Playing, this influence is even negative (User_Score coefficient = -0.1183, $p = 0.008$).

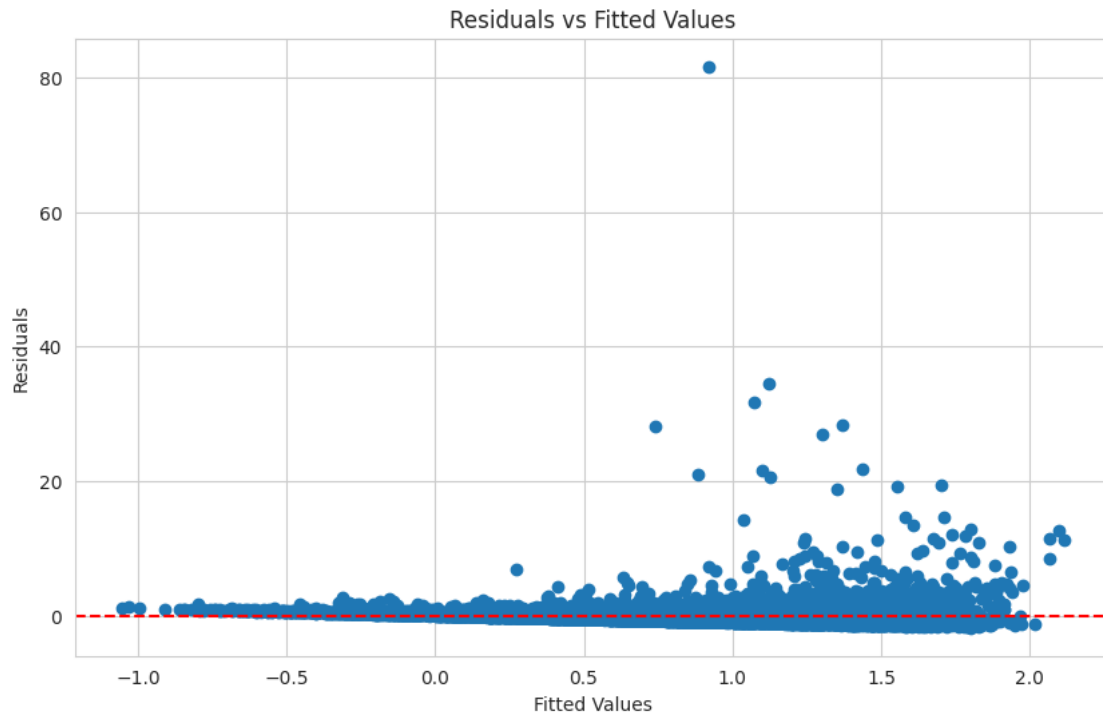
The significance and impact of ratings vary by genre, which may reflect differences in audience and critics' expectations for different types of games.

###Additional analysis After fitting a model, the following assumptions should not be violated:

1. Variability of the residuals is nearly constant. 2. Residuals of the model are nearly normal. 3. Residuals are independent 4. Each variable is linearly related to the outcome.

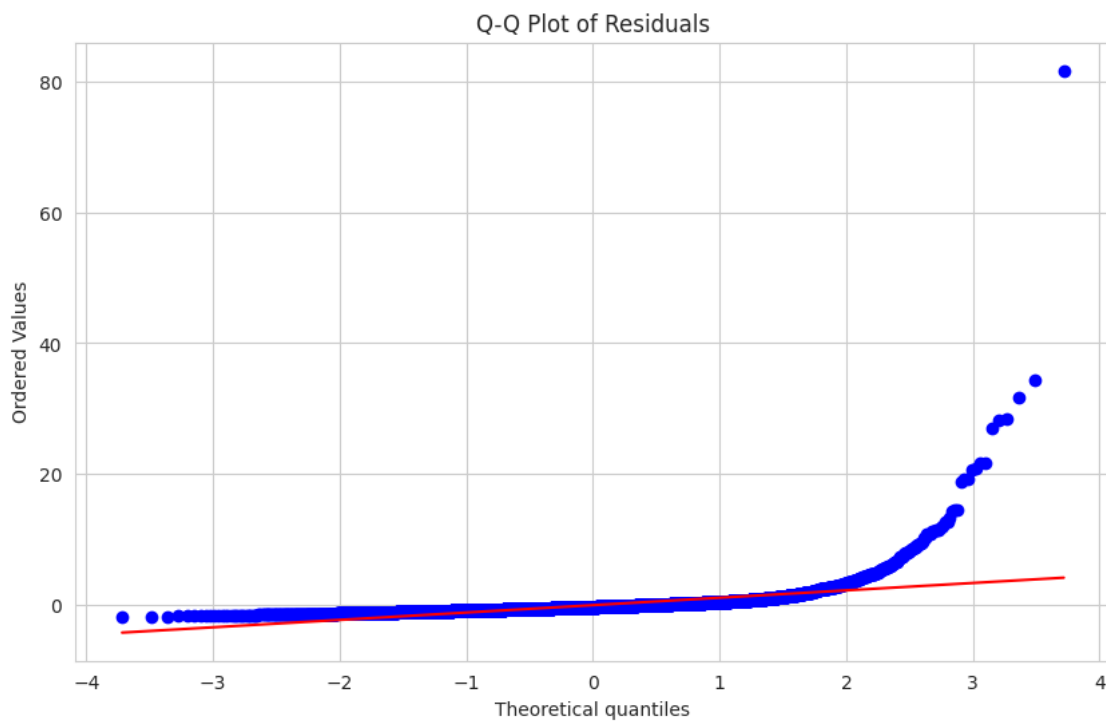
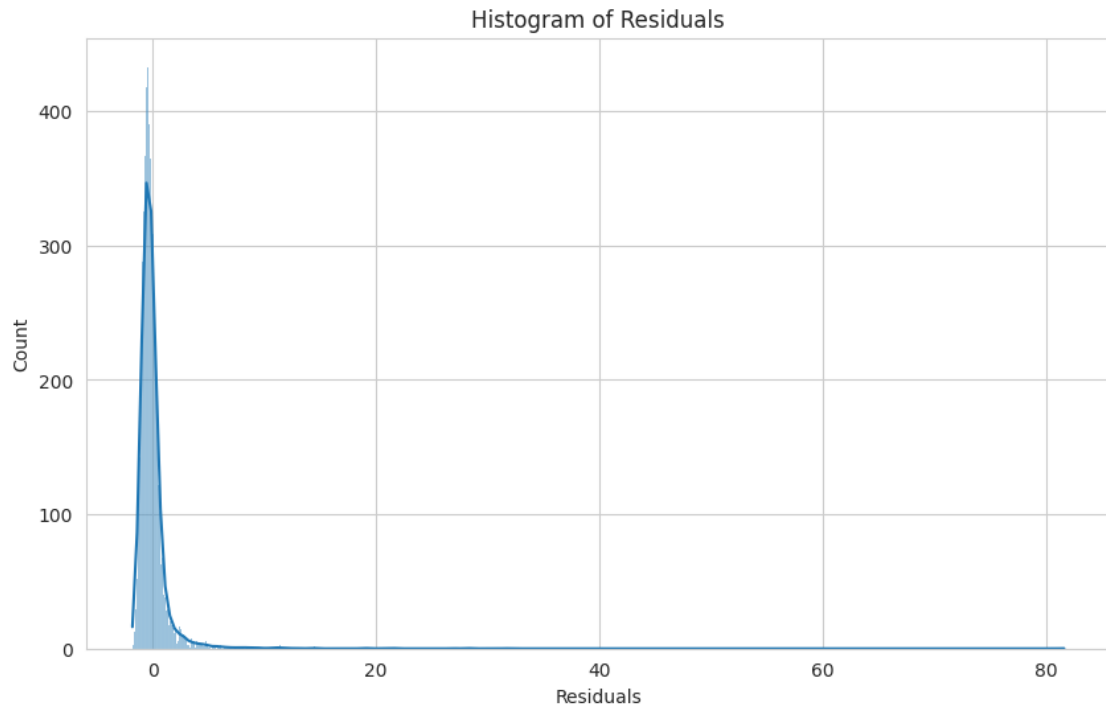
To ensure the assumptions are met, scatterplots, histograms and QQ plots are used to validate the variables selected.

```
[13]: # 1. Check for Homoscedasticity
plt.figure(figsize=(10, 6))
plt.scatter(final_model.fittedvalues, final_model.resid)
plt.axhline(0, color='r', linestyle='--')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residuals vs Fitted Values')
plt.show()
```



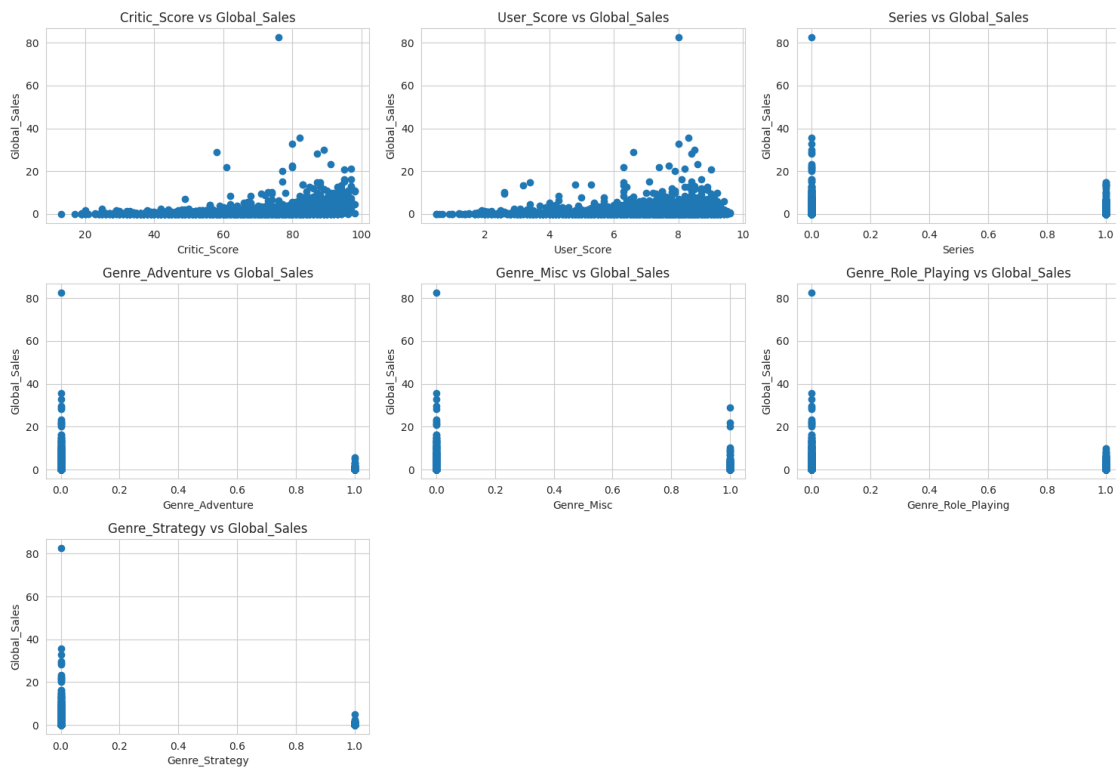
```
[14]: # 2. Check for Normality of Residuals
plt.figure(figsize=(10, 6))
sns.histplot(final_model.resid, kde=True)
plt.xlabel('Residuals')
plt.title('Histogram of Residuals')
plt.show()

# Q-Q plot
plt.figure(figsize=(10, 6))
probplot(final_model.resid, dist="norm", plot=plt)
plt.title('Q-Q Plot of Residuals')
plt.show()
```



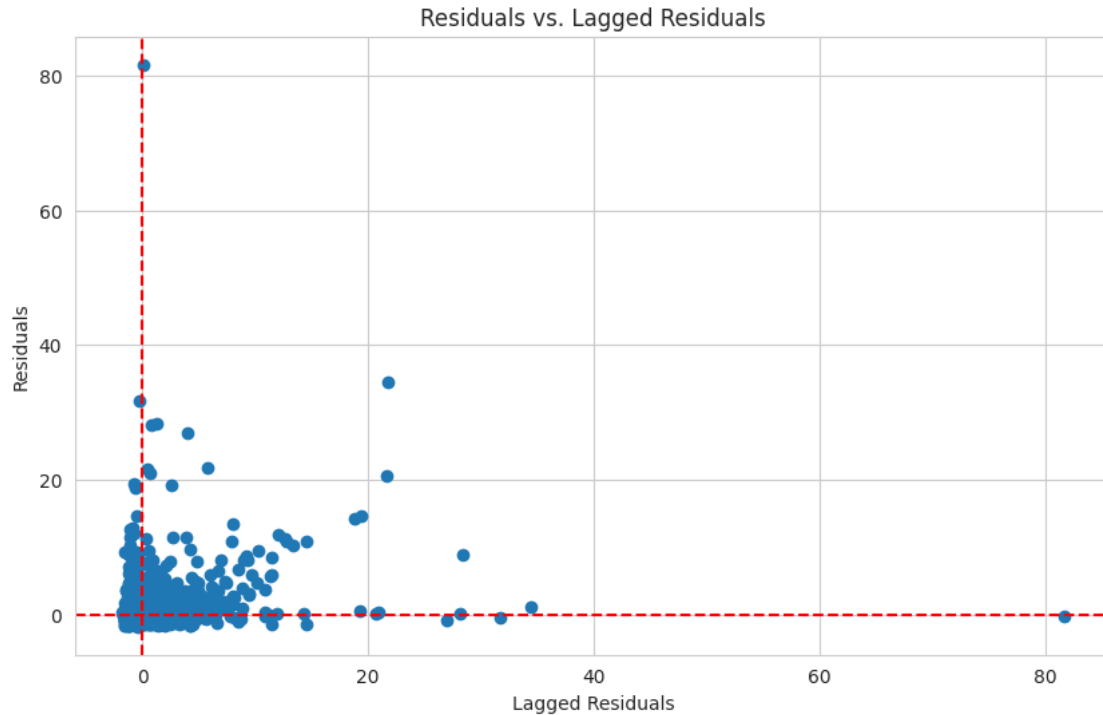
```
[15]: # 3. Check for Linearity
# Scatter plot for each predictor against the outcome
predictors = ['Critic_Score', 'User_Score', 'Series'] + \
    [col for col in games_encoded.columns if col.startswith('Genre_')
    and col != 'Genre_Sports' and col != 'Genre_Puzzle' and col !=
    'Genre_Simulation' and col != 'Genre_Fighting' and col != 'Genre_Racing' and
    col != 'Genre_Shooter' and col != 'Genre_Platform']

plt.figure(figsize=(15, 20))
for i, predictor in enumerate(predictors, 1):
    plt.subplot(6, 3, i)
    plt.scatter(games_encoded[predictor], games_encoded['Global_Sales'])
    plt.xlabel(predictor)
    plt.ylabel('Global_Sales')
    plt.title(f'{predictor} vs Global_Sales')
plt.tight_layout()
plt.show()
```



```
[16]: # 4. Check for Independence of Residuals
# Plot residuals vs. lagged residuals
plt.figure(figsize=(10, 6))
plt.scatter(final_model.resid[:-1], final_model.resid[1:])
plt.xlabel('Lagged Residuals')
```

```
plt.ylabel('Residuals')
plt.title('Residuals vs. Lagged Residuals')
plt.axhline(0, color='r', linestyle='--')
plt.axvline(0, color='r', linestyle='--')
plt.show()
```



The scatter plot of Predicted vs. Absolute Residuals shows that the variability of the residuals is nearly constant. Both the QQ-plot and the histogram of the residuals show that they are distributed normally as they follow the line of the QQ-plot and its distribution is bell shaped but right-skewed. The residuals are independent. The rest of the scatter plots suggests that there is a linear relationship between the outcome and each predictor. As all assumptions are not violated, the fitted model does not require any further improvements such as variable transformation or seeking out additional variables to fill model gaps.

##Results The results of this study showed that many factors have a significant impact on global video game sales. In this article, we have built a multiple linear regression model to determine how criteria such as user and critic ratings, the genre of the game, whether the game is part of a series, as well as attention to the plot and gameplay are related to global game sales. The multiple linear regression model showed the following key results:

$$\hat{y} = 0 + 1 \times \text{Genre_Adventure}[T.True] + 2 \times \text{Genre_Misc}[T.True] + 3 \times \text{Genre_Role_Playing}[T.True] + 4 \times \text{Genre_Strategy}[T.True] + 5 \times \text{Critic_Score}$$

where:

$$0 \text{ (Intercept)} = -1.2665$$

$$1 \text{ (Genre_Adventure}[T.True]) = -0.3435$$

2 (Genre_Misc[T.True]) = 0.3835
3 (Genre_Role_Playing[T.True]) = -0.1985
4 (Genre_Strategy[T.True]) = -0.6227
5 (Critic_Score) = 0.0381

Influencing factors: critical evaluation (Critic_Score) and belonging to the series have a significant positive impact on global game sales user ratings (User_Score) have a negative impact on sales. These results confirm the hypothesis about the impact of evaluation and serialization on the success of the game.

R-square: The value of the R-square (R^2) is 0.071, which means that about 7.1% of the change in global sales is due to the variables used. This reflects the limited ability of the model to explain all the variability of the data, indicating the importance of other factors in addition to the factors considered.

With the exception of the Critic_Score and Series indicators, other variables such as Story_Focus and Gameplay_Focus do not have a statistically significant impact on global sales. This shows that, at least in the context of this model, focusing on the plot and gameplay is not a decisive factor for selling the game on the global market.

Graphical visualization of data, such as histograms of the distribution of global sales, diagrams of the dependence of global sales on the ratings of critics and users, as well as histograms of average global sales by genre, complement the analysis by providing a visual representation of the relationships between variables.

Thus, our research has confirmed that critical acclaim and a series of games have a significant impact on the commercial success of the game, but user ratings have a significant impact on the commercial success of the game.

##Conclusion Using the constructed model, which is given in the “Results” section, we can answer the research question: which of the predictors (critic’s assessment, user assessment, game focus, series, genre) affect further global sales, and also to what extent? According to the model, critics’ ratings and series strongly and positively correlate with sales. At the same time, user ratings correlate significantly, but negatively. It is noteworthy that, unlike the expectations of the researchers outlined in the introduction, there was no evidence that the genre has an impact on its sales, as well as what the focus of this game is. Now this data can be used in different conditions to calculate the level of global sales of games and this will make the gaming industry even better. ###Limitations During the execution of this study, certain limitations were identified that must be taken into account for the correct interpretation of the results:

The study is based on data collected in 2016. Consequently, the latest trends and changes in the gaming industry after this period could remain unnoticed, which may reduce the applicability of the model in modern conditions. In the future, it is advisable to re-analyze from the very beginning, starting with data collection.

The low value of the R^2 determination coefficient of the model (approximately 0.071) indicates that the proportion of variability in global video game sales explained by the model is insignificant. This indicates that there are many other factors that are not taken into account in the model and can have a significant impact on global video game sales.

##References [The final data set \(from kaggle.com\)](#)

[The original data set \(by Timo Potanen\)](#)